YUXIANG MA

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ABOUT ME

PhD candidate at MIT working on visuo-tactile perception and robot learning for contact-rich manipulation. My work runs from tactile sensor and gripper design, through human and robot data collection, to video world models and foundation-model systems deployed on real robots.

EDUCATION

Massachusetts Institute of Technology, Cambridge, MA *Sep. 2021–Present*
PhD and S.M. (2023), Mechanical Engineering. Advisor: Edward Adelson. GPA: 5.0/5.0

Tsinghua University, Beijing *Aug. 2016–Jul. 2020*
B.Eng., Engineering Mechanics, Tsien Excellence in Engineering Program. GPA: 3.85/4.0, rank 2/28

WORK EXPERIENCE

Research Intern, Robotics and AI Institute, Cambridge, MA *Jun. 2025–Aug. 2025*
Worked with Team Capture on portable data collection for robot learning: built a handheld UMI-style device with force-torque and tactile sensors, and the synchronized software that records visuo-tactile manipulation data.

RESEARCH

CoStream: Composing Simple Behaviors for Complex Manipulation *Sep. 2025–Feb. 2026*
GelSight Lab, MIT & Embodied Minds Lab, Harvard *CoRL 2026, co-first author*
Split manipulation into independent behaviors: spatial constraints from LLMs/VLMs, trajectories from a video world model, and high-rate tactile and force corrections. Each outputs an SE(3) target, and the targets compose into one pose command per control step. Solves 8 real-world tasks, including CPU, GPU, and RAM insertion, with no per-task demonstrations or retraining.

ACTS: Visuo-Tactile World Model Learned from Human Physical Interaction *2026*
GelSight Lab, MIT *arXiv 2026, first author*
Built React, a 4.9-hour bimanual human dataset from handheld GelSight sensors (scene, wrist, and tactile video, 6-DoF motion, normal force). Adapted a Cosmos video generator with LoRA to jointly predict five visual and tactile streams from force-aware actions: 0.7–2.0 dB higher scene PSNR over visuo-tactile baselines, with zero-shot tactile transfer to robot teleoperation.

Tactile 6-DoF Object Pose Estimation (YOTO, InvariantCloud) *2025–2026*
GelSight Lab, MIT, with HKUST and University of Cambridge *ICRA 2026 and preprint, co-first author*
YOTO recovers 6-DoF object pose from one pair of tactile contacts: a network pretrained on virtual tactile patches localizes each contact on a CAD or phone-scanned mesh, and a closed-form SVD solver returns the pose. It stays accurate under occlusion, where FoundationPose drifts. InvariantCloud tracks pose from globally invariant marker point clouds, reducing yaw drift.

Tactile Sensors and Fingers (GlowTact, GelLink, GelSight Fin Ray) *Oct. 2021–2026*
GelSight Lab, MIT *ICRA 2024, RoboSoft 2024, and preprint; first or co-first author*
GlowTact: a compact sensor using only single-color light and a monochrome camera, with no photometric stereo or calibration; detects a 0.2 g screw and resolves 0.25 mm threads. GelLink: a three-phalanx, single-motor finger that uses one camera for both touch and proprioception. Fin Ray: FEM and rendering simulation for designing compliant tactile fingers.

PUBLICATIONS (* equal contribution)

2026

CoRL CoStream: Composing Simple Behaviors for Generalizable Complex Manipulation
H. Chen*, **Y. Ma***, S. Tian, X. Han, W. Huang, F. Wu, Y. Li, J. Wu, E. Adelson, Y. Du [website]

ICRA InvariantCloud: A Globally Invariant, Uniquely Indexed Point Cloud Framework for Robust 6-DoF Tactile Pose Tracking
P. Ye*, **Y. Ma***, Y. Zhou, W. Chen, W. Dong, M. Duan [arXiv]

WAFR RichMap: A Reachability Map Balancing Precision, Efficiency, and Flexibility for Rich Robot Manipulation Tasks
Y. Lu, **Y. Ma**, J. Pan [arXiv] [code]

arXiv ACTS: Learning Visuo-Tactile World Models from Human Physical Interaction
Y. Ma*, W. Xu*, P. Ye, A. Mei, Y. R. Song, T. Li, H. Chen, E. Adelson [arXiv]

arXiv Robot Learning with Visual Predicted Force
H. Chen*, F. Wu*, **Y. Ma**, M. Mete, P. Ye, . . . , R. D. Howe, Y. Du [website]

arXiv GlowTact: Simple and Compact Vision-Based Tactile Sensing with High Sensitivity and Spatial Resolution
Y. Ma*, M. Tippur*, P. Ye, S. Q. Liu, H. Chen, F. R. Cottrell, E. Adelson [website]

arXiv A Brain-inspired Hierarchical Framework for Zero-Shot Robot Task Reasoning and Execution
G. Wang, P. Ye, Q. Ying, Y. Jing, **Y. Ma**, H. Chen, . . . , M. Duan, B. Sheil [arXiv]

arXiv RoboEdit: Turning Human Manipulation Videos into Scalable Robot Experience
Y. Guo, Z. Tao, Y. Jiang, Y. Chen, Z. Dou, **Y. Ma**, Y. Yang, . . . , C. Jiang [website]

arXiv You Only Touch Once: 6-DoF Object Pose Estimation from Single Tactile Contact
P. Ye*, **Y. Ma***, H. Chen, G. Wang, Y. Jing, B. Sheil, Y. Du, E. Adelson [arXiv]

2025

ICRA-W Dynamic Regrasping with Asynchronous Vision Feedback using a Minimalist Robotic System
Y. Ma, E. Adelson *RoboLetics 2.0 Workshop* [paper]

2024

CoRL Transferable Tactile Transformers for Representation Learning Across Diverse Sensors and Tasks
J. Zhao, **Y. Ma**, L. Wang, E. Adelson [website]

ICRA GelLink: A Compact Multi-phalanx Finger with Vision-based Tactile Sensing and Proprioception
Y. Ma, J. Zhao, E. Adelson [arXiv]

RoboSoft Scalable, Simulation-Guided Compliant Tactile Finger Design
Y. Ma*, A. Agarwal*, S. Q. Liu*, W. Yuan, E. Adelson [arXiv]

2023

Nat. BME Stretchable Ultrasonic Arrays for the Three-Dimensional Mapping of the Modulus of Deep Tissue
H. Hu*, **Y. Ma***, X. Gao*, D. Song*, . . . , S. Xu *Nature Biomedical Engineering* [paper]

RoboSoft GelSight Baby Fin Ray: A Compact, Compliant, Flexible Finger with High-Resolution Tactile Sensing
S. Q. Liu, **Y. Ma**, E. Adelson [arXiv]

Nature A Wearable Cardiac Ultrasound Imager
H. Hu, H. Huang, M. Li, X. Gao, L. Yin, R. Qi, R. S. Wu, X. Chen, **Y. Ma**, . . . , S. Xu [paper]

CoRL-W Object Recognition and Force Estimation with the GelSight Baby Fin Ray
S. Q. Liu, **Y. Ma**, E. Adelson *Workshop on Learning for Soft Robots* [arXiv]

2021

EML The Effect of Arterial Stiffness on the Accuracy of Cuff-Based Blood Pressure Measurement
Y. Ma, Y. Chen, Y. Ma, X. Feng *Extreme Mechanics Letters* [paper]

SKILLS

Programming: Python, C/C++, ROS
Learning: video world models, tactile representation learning, LoRA fine-tuning
Hardware/simulation: SolidWorks, Onshape, ABAQUS, Mitsuba, Blender, Unity, machine shop